

Chapter III

Detailed Solutions

1 INPhO-2006

1.

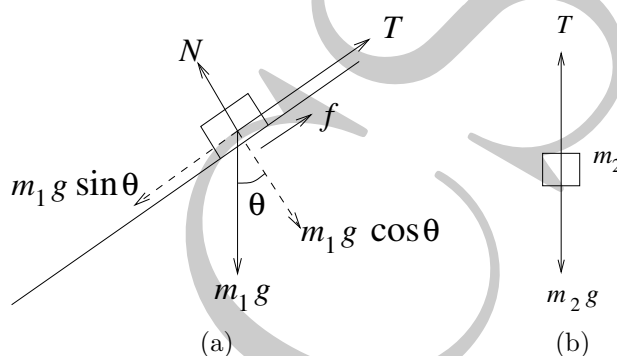


Figure 1: Free body diagrams assuming m_1 is going down: (a) For m_1 . (b) For m_2 .

(a) Let m_1 go down. Then from free body diagram for m_1 ,

$$m_1 g \sin \theta - f - T = m_1 a \quad (1)$$

where the frictional force $f = \mu N$, N being the normal reaction. Hence,

$$m_1 g \sin \theta - \mu N - T = m_1 a$$

$$N = m_1 g \cos \theta$$

$$\text{Therefore} \quad a = g \sin \theta - \mu g \cos \theta - \frac{T}{m_1} \quad (2)$$

From free body diagram for m_2 ,

$$T - m_2 g = m_2 a \quad (3)$$

Solving Eqs. (2) and (3),

$$a = \frac{g [m_1 (\sin \theta - \mu \cos \theta) - m_2]}{m_2 + m_1}$$

As $m_1 = m_2$,

$$a = \frac{g [\sin \theta - 1 - \mu \cos \theta]}{2}$$

We can see from the above equation that a can have only negative values. Substituting the values

$$a = -5.1 \text{ m}\cdot\text{s}^{-2} < 0$$

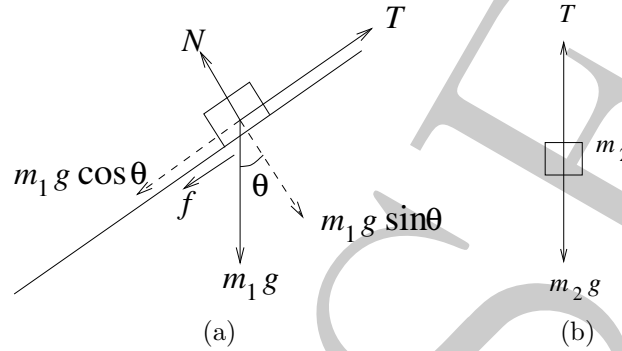


Figure 2: Free body diagrams assuming m_2 is going down: (a) For m_1 . (b) For m_2 .

Let m_2 go down. Then for m_2

$$m_2 g - T = m_2 a \quad (4)$$

For m_1

$$T - m_1 g \sin \theta - f = m_1 a \quad (5)$$

Elementary algebraic manipulations yield

$$a = \frac{g [1 - \sin \theta - \mu \cos \theta]}{2}$$

Substituting the values

$$a = -0.1 \text{ m}\cdot\text{s}^{-2} < 0$$

As seen from both the cases, the acceleration is negative, either assuming m_1 going down or m_2 going down. So we can conclude that the acceleration of the system is zero and **system would be stationary**.

(b) substituting $a = 0$ in Eq. (4),

$$T \cong 10 \text{ N.}$$

Since system is stationary, so frictional force would be static friction and a may have a positive value only if m_2 is going down and not up. By substituting the value of T and a in Eq. (5), we obtain

$$f_s = 5.2 \text{ N.}$$

This f_s is down the inclined plane. Why?

2.

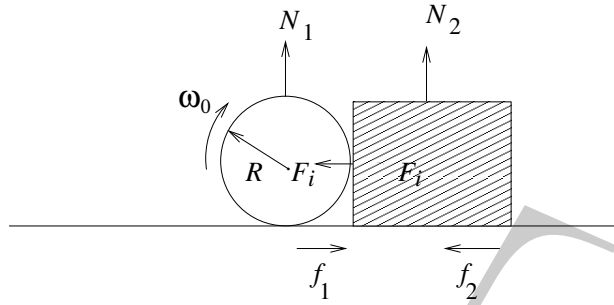


Figure 3: Free body diagram of the block disk system

Key points to remember are, when disk touches the block, block-disk system starts to slide. After some time t^* , disk starts to roll without slipping and after this time t^* , disk continues to roll without slipping and comes to a halt in total time t_{tot} .

- (a) If a is the acceleration of the block-disk system, then by free body diagram(see Fig. (3)).

$$\begin{aligned} (2M + M)a &= f_1 - f_2 \\ 3Ma &= \mu(2M)g - \mu Mg \\ a &= \frac{\mu g}{3} \end{aligned} \quad (6)$$

- (b) As disk rolls, it's angular velocity ω decreases. If α is the angular acceleration of the system, then

$$\begin{aligned} I\alpha &= -f_1 R \\ \frac{1}{2}(2M)R^2\alpha &= -\mu(2M)g R \\ \alpha &= \frac{-2\mu g}{R} \end{aligned} \quad (7)$$

At time t^* , disk starts to roll without slipping. So

$$v(t^*) = R\omega(t^*) \quad (8)$$

and equation of motion for rotational motion

$$\omega(t^*) = \omega_0 - \frac{2\mu g}{R} t^* \quad (9)$$

and equation of motion for translational motion

$$v(t^*) = 0 + \frac{\mu g}{3} t^* \quad (10)$$

By Eqs. (9) and (10)

$$t^* = \frac{3}{7} \cdot \frac{R\omega_0}{\mu g} \quad (11)$$

- (c) After time $t > t^*$, velocity of block - disk system starts to decrease. In this condition of pure rolling, frictional force on disk $f'_1 < \mu(2M)g$. Suppose angular acceleration is α' and linear acceleration is a' , then

$$\alpha' = \frac{a'}{R}$$

$$I \alpha' = -f'_1 R$$

$$\frac{1}{2}(2M)R^2 \alpha' = -f'_1 R$$

Hence

$$-f'_1 = M a' \quad (12)$$

Now

$$(2M + M)a' = f'_1 - f'_2$$

$$3M a' = -M a' - \mu M g \Rightarrow a' = \frac{-\mu g}{4} \quad (13)$$

Suppose in time t from t^* the system comes to rest. Then

$$v(t) = v(t^*) + a't$$

Using Eqs. (10) and (13)

$$\begin{aligned} 0 &= \frac{\mu g}{3} \cdot \frac{3R\omega_0}{7\mu g} - \frac{\mu g}{4} t \\ &= \frac{R\omega_0}{7} - \frac{\mu g t}{4} \\ t &= \frac{R\omega_0}{\mu g} \cdot \frac{4}{7} \end{aligned}$$

Thus total time

$$\begin{aligned} t_{tot} &= t + t^* \\ t_{tot} &= \frac{R\omega_0}{\mu g} \end{aligned}$$

This is a surprisingly simple answer.

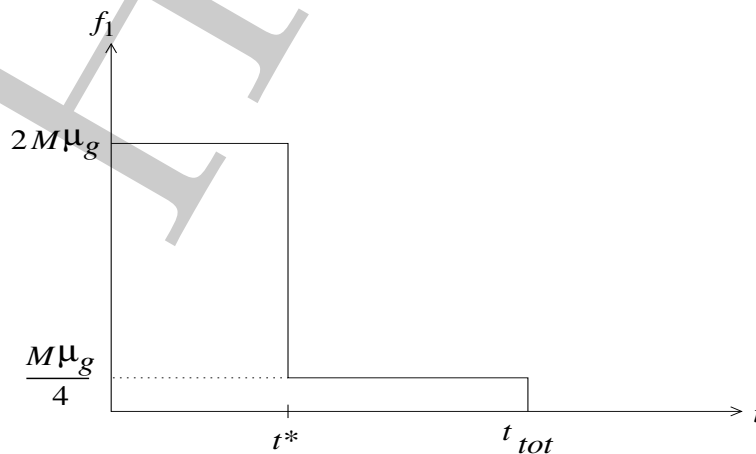


Figure 4: Plot of force of friction with time

Figure (4) gives the interesting plot of force of friction (f_1) with time. One can make an equivalent plot with angular velocity ω on the x-axis. There is a sudden drop in f , at $t = t^* = 3r\omega_0/7\mu g$. This drop occurs when pure rolling sets in. It may remind the reader of a first order phase transition where f_1 plays the role of the internal energy and ω that of temperature.

3. (a) Efficiency of cycle is

$$\eta = 1 - \frac{Q_{out}}{Q_{in}}$$

In the cycle, energy is absorbed in isobaric process from a to b . So $Q_{in} = C_p(T_b - T_a)$. Here C_p is the heat capacity at constant pressure. Energy is released in isobaric process c to d . So $Q_{out} = C_p(T_c - T_d)$. Hence

$$\eta = 1 - \frac{T_c - T_d}{T_b - T_a} \quad (14)$$

Another method exists for this solution. In this method

$$\eta = \frac{W}{Q_{in}}$$

and W is evaluated by integrating the area under the $P - V$ curve.

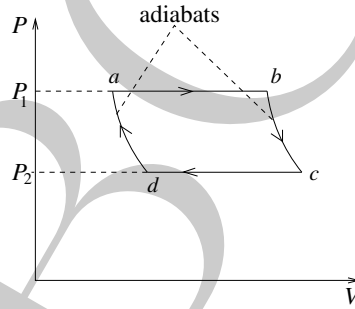


Figure 5: P - V diagram

- (b) For adiabats b to c and d to a

$$PV^\gamma = \text{const.} \quad (15)$$

Equation of state gives

$$PV = nRT \Rightarrow \frac{PV}{T} = \text{const.}$$

Inserting this in Eq. (15)

$$T = \text{const.} \times P^{\frac{\gamma-1}{\gamma}} \quad (16)$$

Equation (16) gives

$$\frac{T_c}{T_d} = \frac{T_b}{T_a} \quad (17)$$

and

$$\frac{T_d}{T_a} = \left(\frac{P_2}{P_1} \right)^{\frac{\gamma-1}{\gamma}}$$

Equation (14) becomes

$$\eta = 1 - \left(\frac{P_2}{P_1} \right)^{\frac{\gamma-1}{\gamma}}$$

- (c) For isobaric process, $V - T$ curve will be a straight line passing through the origin of the $V - T$ plot. Since $P_1 > P_2$ so for $a - b$, this line will be less steeper than $c - d$. By Eq. (15) and equation of state, we can get $T_b > T_c > T_a > T_d$. For convexity of curve from $a - d$ and $b - c$, we can analyze relation between V and T , that slope is inversely proportional to T . So on this basis we can draw $V - T$ diagram as shown in Fig. (6).

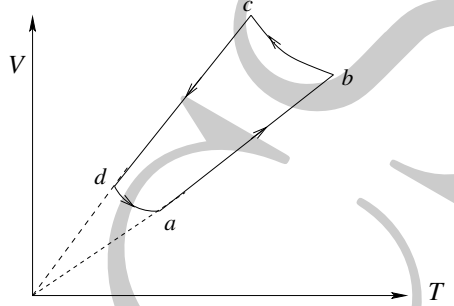


Figure 6: $V-T$ diagram

- (d) Carnot efficiency is defined by the highest temperature T_b and lowest temperature T_d :

$$\eta_c = 1 - \frac{T_d}{T_b}$$

Efficiency of concerned cycle (from Eqs. (14) and (17)) is

$$\eta = 1 - \frac{T_d}{T_a}$$

Hence

$$\frac{\eta_c}{\eta} = \frac{1 - \frac{T_d}{T_b}}{1 - \frac{T_d}{T_a}}$$

Since $T_b > T_a$, hence $\eta_c > \eta$ i.e. Carnot engine has higher efficiency than the concerned cycle.

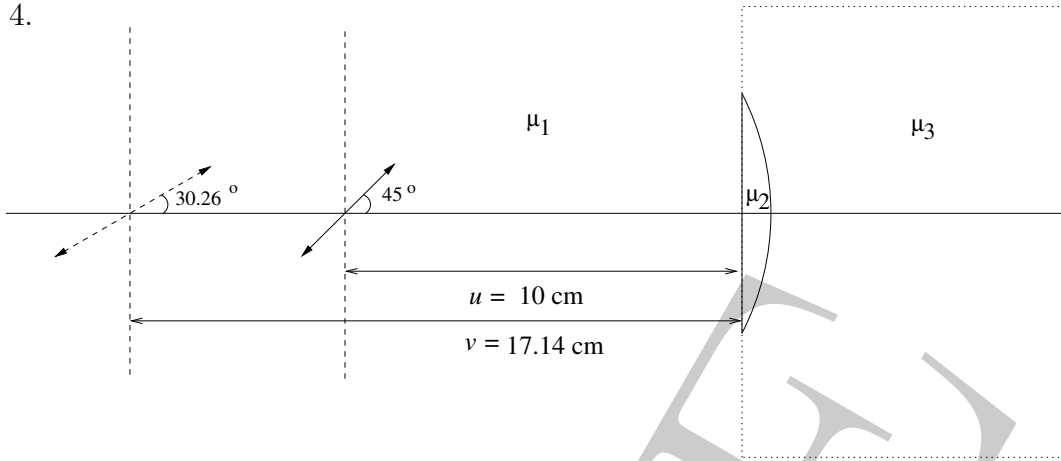


Figure 7: Problem 4

Given $\mu_1 = 1.0$, $\mu_2 = 1.5$, $\mu_3 = 1.2$, $u = -10$ cm, $R = -10$ cm.

General formula for thin lens,

$$\frac{\mu_3}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R_1} - \frac{\mu_2 - \mu_3}{R_2}$$

Now $R_1 \rightarrow \infty$, $R_2 = R$.

$$\text{Therefore} \quad \frac{\mu_3}{v} - \frac{\mu_1}{u} = \frac{\mu_3 - \mu_2}{R} \quad (18)$$

$$\begin{aligned} \text{Implies} \quad v &= \frac{Ru\mu_3}{u(\mu_3 - \mu_2) + \mu_1 R} \\ &= -17.14 \text{ cm} \end{aligned} \quad (19)$$

Here the negative sign indicates that image is on the left side. Under small shift along the principal axis of the lens for the object, the image will also shift slightly along the same axis. So, for longitudinal magnification differentiate Eq. (19) with respect to v and get

$$\begin{aligned} \frac{dv}{du} &= \left(\frac{v}{u}\right)^2 \frac{\mu_1}{\mu_3} \\ &= 2.4 \end{aligned}$$

During the first refraction, ray travels from medium 1 (μ_1) to medium 2 (μ_2). Linear lateral magnification in first refraction

$$\frac{y'_2}{y_1} = \frac{\mu_1}{\mu_2} \frac{v'}{u} \quad (20)$$

For the second refraction

$$\frac{y_2}{y'_2} = \frac{\mu_2}{\mu_3} \frac{v}{v'} \quad (21)$$

Eq. (20) and Eq. (21) gives

$$\begin{aligned} \frac{y_2}{y_1} &= \frac{\mu_1}{\mu_3} \frac{v}{u} \\ &= 1.43 \end{aligned}$$

It is given that object's amplitude is $\sqrt{2}$ cm. Taking the projection of object along the principle axis and perpendicular to the axis,

$$\begin{aligned} du &= 1 \text{ cm (along the axis)} \\ \text{and } y_1 &= 1 \text{ cm (normal to the axis)} \\ \Rightarrow dv &= 2.4 \text{ cm} \\ \text{and } y_2 &= 1.4 \text{ cm.} \end{aligned}$$

$$\begin{aligned} \text{Amplitude of the image} &= \sqrt{(dv)^2 + y_2^2} \\ &= 2.78 \text{ cm} \end{aligned}$$

Since lateral and longitudinal magnifications are positive, the phase difference between oscillating image and object will be zero. Only orientation of the axis along which oscillation takes place will be different.

Angle made by the virtual image with the principle axis

$$\begin{aligned} &= \tan^{-1} \left(\frac{y_2}{dv} \right) \\ &= \tan^{-1} \left(\frac{1.4}{2.4} \right) \\ &= 30.26^\circ \end{aligned}$$

5.

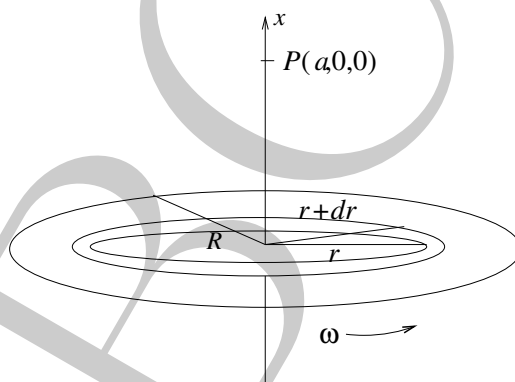


Figure 8: Problem 5

(a) Dipole moment due to a circular ring $(r, r + dr)$ (See Fig. (8))

$$\begin{aligned} d\mu' &= di A \\ \text{And } di &= \frac{dQ}{T} = \frac{2\pi r dr \sigma}{2\pi/\omega} \\ d\mu' &= \frac{2\pi r dr \omega \sigma}{2\pi} \cdot \pi r^2 \end{aligned} \tag{22}$$

Then for circular disk

$$\begin{aligned} \mu' &= \pi\omega\sigma \int_0^R r^3 dr \\ \mu' &= \frac{\pi\omega\sigma R^4}{4} \end{aligned}$$

Direction is in the positive x -direction.

(b) Magnetic field due to a circular ring ($r, r + dr$)

$$dB = \frac{\mu_0}{2} \cdot \frac{di r^2}{(r^2 + a^2)^{3/2}}$$

$$B = \frac{\mu_0}{2} \int_0^R \frac{di r^2}{(r^2 + a^2)^{3/2}}$$

Using Eq. (22)

$$B = \frac{\mu_0 \omega \sigma}{2} \int_0^R \frac{r^3}{(r^2 + a^2)^{3/2}} dr$$

Substitutions

$$r = a \tan \theta$$

$$dr = a \sec^2 \theta d\theta$$

and limit 0 to $\tan^{-1}(\frac{R}{a})$

$$B = \frac{\mu_0 \omega \sigma a}{2} \int_0^{\tan^{-1}(\frac{R}{a})} \tan^2 \theta \sin \theta d\theta$$

$$= \frac{\mu_0 \omega \sigma a}{2} \int_0^{\tan^{-1}(\frac{R}{a})} (\sec^2 \theta - 1) \sin \theta d\theta$$

$$= \frac{\mu_0 \omega \sigma a}{2} \int_0^{\tan^{-1}(\frac{R}{a})} (\sec^2 \theta \sin \theta - \sin \theta) d\theta$$

Integrating the first term by parts

$$B = \frac{\mu_0 \omega \sigma a}{2} [\sin \theta \tan \theta + 2 \cos \theta]_0^{\tan^{-1}(\frac{R}{a})}$$

Applying the limits

$$B = \frac{\mu_0 \omega \sigma}{2} \left[\frac{R^2 + 2a^2}{\sqrt{R^2 + a^2}} - 2a \right] \quad (23)$$

Direction is in the positive x -direction.

(c) **Method - I**

When $a \gg R$ then for axial field due to dipole, we can use the formula

$$B = \frac{\mu_0}{4\pi} \cdot \frac{2\mu'}{a^3}$$

$$= \frac{\mu_0}{4\pi} \cdot \frac{2}{a^3} \cdot \frac{\pi \sigma \omega R^4}{4}$$

$$= \frac{\mu_0 \sigma \omega R^4}{8a^3}$$

Direction is in the positive x -direction.

Method - II

From Eq. (23)

$$\begin{aligned}
 B &= \frac{\mu_0 \omega \sigma}{2} \left[\frac{R^2 + 2a^2}{\sqrt{R^2 + a^2}} - 2a \right] \\
 &= \frac{\mu_0 \omega \sigma}{2} \left[\frac{R^2 + 2a^2}{a} \left(1 + \frac{R^2}{a^2} \right)^{-\frac{1}{2}} - 2a \right] \\
 &= \frac{\mu_0 \omega \sigma}{2} \left[\left(\frac{R^2}{a} + 2a \right) \left(1 - \frac{R^2}{2a^2} + \frac{3R^4}{8a^4} \right) - 2a \right] \\
 &= \frac{\mu_0 \omega \sigma R^4}{8a^3}
 \end{aligned} \tag{24}$$

Direction is in the positive x -direction.

(d) The force on dipole of dipole moment μ placed at a

$$F = \mu \frac{dB}{dx}$$

After differentiating Eq. (24) with respect to a ,

$$F = -\frac{3\mu\mu_0\omega\sigma R^4}{8a^4}$$

Direction is dictated by $\vec{\mu}$.

6.

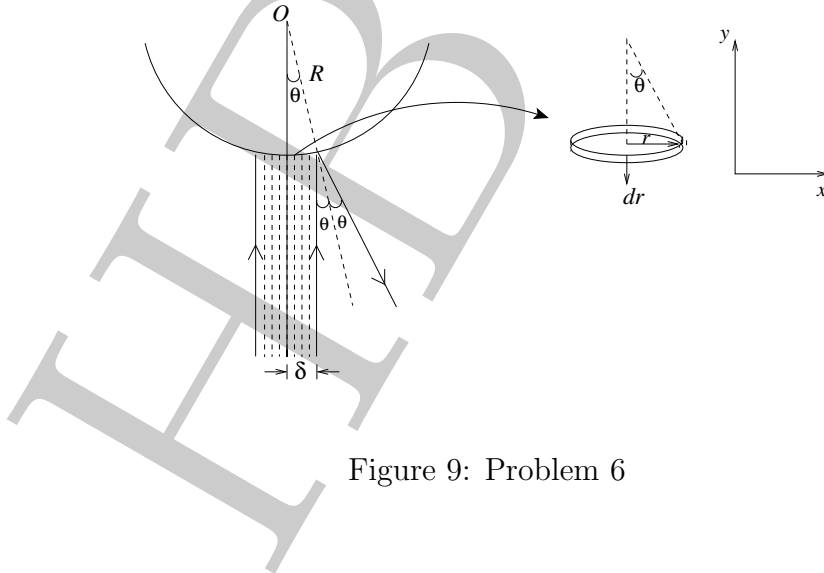


Figure 9: Problem 6

- (a) Initial momentum of photon in the beam $\vec{p}_i = p\hat{j}$.
 Final momentum of photon in the beam $\vec{p}_f = p \sin 2\theta \hat{i} - p \cos 2\theta \hat{j}$.
 x component of momentum cancels by symmetry.
 Net change in momentum $\Delta\vec{p} = -2p \cos^2 \theta$.
 The number of photons in the annular region r to $r + \Delta r$ per second
 $= n 2\pi r dr$.
 Magnitude of force on the annular region is $\Delta F = n 2\pi r dr \times 2p \cos^2 \theta$.

Total force on the sphere is (magnitude)

$$\begin{aligned}
 F &= \int_0^\delta 4\pi p r \cos^2 \theta dr \\
 \sin \theta &= \frac{r}{R} \quad \cos^2 \theta = 1 - \frac{r^2}{R^2} \\
 F &= 4\pi np \left[\frac{\delta^2}{2} - \frac{\delta^4}{4R^2} \right] \quad (25)
 \end{aligned}$$

Direction is upward so $\vec{F} = F\hat{j}$.

(b) **Method - I**

Since $\delta \ll R$, drop δ^4/R^2 term in previous part.

$$F \approx 4\pi np \frac{\delta^2}{2} \quad (26)$$

If E (or $h\nu$) is the energy of one photon and P the power of the beam (1 kW), then

$$\frac{P}{E} = n\pi\delta^2 \quad (27)$$

$$\text{Also } \frac{P}{E} = pc \quad (28)$$

Using Eqs. (27) and (28) in Eq. (26)

$$F = 2 \frac{P}{c}$$

For levitation

$$\begin{aligned}
 F &= mg \\
 m &= \frac{2P}{gc} = 6.80 \times 10^{-7} \text{ kg.}
 \end{aligned}$$

Method - II

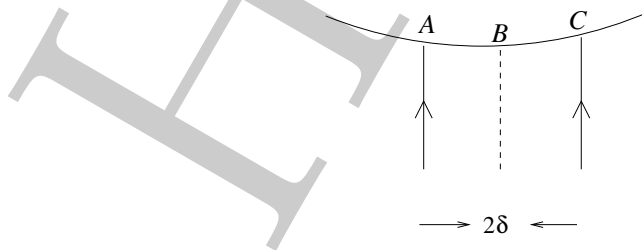


Figure 10: Problem 6 (b) method II

Alternatively one can obtain this without part(a).

Area $= \pi \delta^2$.

Since $\delta \ll R$, we can assume that A-B-C is flat.

Change in momentum per photon $= 2p$.

No. of photons per unit area per sec. = n

Therefore

$$\begin{aligned} F &= (n \pi \delta^2) 2p \\ &= 2 \pi n p \delta^2 \end{aligned}$$

This is the same as Eq. (26) and we proceed as in the previous part.

Although the answer to the question ends here, we draw your attention to the following. Let us estimate the size of this ‘Al’ sphere.

$$\begin{aligned} m &= \frac{4\pi}{3} r^3 \rho \\ r^3 &= \frac{3m}{4\pi\rho} \\ &= \frac{6.8 \times 10^{-7}}{4 \times 2.7 \times 10^3} \times \frac{3}{3.14} \\ &= \frac{6.8 \times 3}{4 \times 2.7 \times 3.14} \times 10^{-10} \\ r &= 10^{-3} \times \left(\frac{6.8 \times 0.3}{4 \times 2.7 \times 3.14} \right)^{1/3} \\ &= 4 \times 10^{-4} \text{ m} = 0.4 \text{ mm}. \end{aligned}$$

Note that focussing a laser beam to a spot size $r/10 = 4 \times 10^{-5}$ m is easy. In this connection, recall that the “minimum” spot size is dictated by the wavelength of the beam. Laser levitations of polystyrene beads have been demonstrated. Optical tweezers is also a possibility. Thus it is possible to realise the experiment.

7. (a) Using formula

$$E_{n_2} - E_{n_1} = 13.6 Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \text{ eV}$$

For Li^{++} , $Z=3$, and transition from $n_2 = 4$ to $n_1 = 3$.

$$E_4 - E_3 = \Delta E = h\nu = h \frac{c}{\lambda} = 13.6 \times 9 \left(\frac{1}{3^2} - \frac{1}{4^2} \right) \text{ eV}$$

Putting the values of h and c ,

$$\lambda = 2089 \text{ \AA}$$

This wavelength belongs to the ultraviolet region of the electromagnetic spectrum.

(b)

$$h\nu = \phi + (\text{K.E.})_{\max}$$

Here ϕ is the work function of metal. Putting all the values,

$$(\text{K.E.})_{\max} = 2.75 \text{ eV}$$

Now de Broglie wavelength

$$\lambda_{dB} = \frac{h}{\sqrt{2 m_e (\text{K.E.})_{\max}}}$$

$$\lambda_{dB} = 7.4 \text{ \AA}$$

8. (a) Equating Energies, (or Pressures, or Forces) for a spherical “charged” cloud

$$\frac{3}{5} \frac{Q^2}{4 \pi \epsilon_0 R} = \frac{3}{5} \frac{G M^2}{R}$$

$$Q = Nq = N(-ey)$$

$$M = N m_p$$

$$y^2 = \frac{m_p^2}{e^2} G 4 \pi \epsilon_0$$

$$y = 0.9 \times 10^{-18} \simeq 10^{-18}$$

Hence

$$y \geq 10^{-18}$$

- (b) The force on the atom at a distance r from the centre is

$$F = \frac{1}{4 \pi \epsilon_0} \frac{Q q}{r^2}$$

$$m \frac{dv}{dt} = m \frac{dv}{dr} \cdot \frac{dr}{dt} = m v \frac{dv}{dr} \quad (\text{chain rule})$$

$$Q = \frac{\eta}{m_p} \frac{4 \pi r^3}{3} q$$

$$q = -e y$$

$$v^2 \propto r^2 \quad \text{or} \quad v \propto r$$

with constant of proportionality

$$H = \left[\frac{\eta}{3 \epsilon_0 m_p^2} e^2 y^2 \right]^{\frac{1}{2}}$$

$$H = \frac{ey}{m_p} \left[\frac{\eta}{3 \epsilon_0} \right]^{\frac{1}{2}}$$

- (c)

$$H = \frac{ey}{m_p} \left[\frac{\eta}{3 \epsilon_0} \right]^{\frac{1}{2}}$$

Putting right values of η , ϵ_0 , m_p , e and $y = 10^{-17}$ (one order of magnitude larger)

$$H = 1.8 \times 10^{-17} \text{ s}^{-1}$$

(d) **Method - I**

$$\begin{aligned}\frac{dr}{dt} &= H r \\ r &= r_0 e^{Ht} \\ V &= V_0 e^{3Ht}\end{aligned}$$

Method - II

$$\begin{aligned}\frac{dV}{dt} &= 3 \cdot \frac{4\pi r^2}{3} \frac{dr}{dt} \\ &= 3 \frac{4\pi r^3}{3} H \\ \frac{dV}{V} &= 3 H dt \\ V &= V_0 e^{3Ht}\end{aligned}$$

- (e) Constant H obtained in part (c) is physically similar to Hubble's constant. Observed value of Hubble's constant is $= 2.3 \times 10^{-18} \text{s}^{-1}$, while obtained H is $= 1.8 \times 10^{-17} \text{s}^{-1}$. Further, experiments do not indicate a difference in the magnitudes of the electron and proton charge. Some theories regarding the nature of the fundamental forces and elementary particles also do not point to a difference.



“This is not right. It is so bad, it is not even wrong!”

-Wolfgang Ernst Pauli (25 Apr.1900-15 Dec.1958)
on the solution to a problem provided by a colleague.